

CS2206 Computer Architecture

Lab 2: Floating-point adder architecture(16bit-IEEE-754 std)

Using basic logic-simulation building blocks, design and implement a floating-point adder. Assume the adder will be used in a 16-bit computer system that follows IEEE-style floating-point arithmetic, where each floating-point number consists of 1 sign bit, 5 exponent bits, and 10 fraction bits. The primary requirement is to support normalized floating-point addition. Additional credit will be awarded for designs that correctly handle special cases.

In Record,

- Clear Block Schematic of your design.
- All special cases associated with the 16-bit floating-point format must be recorded and documented, including the minimum and maximum representable values, zero, as well as examples of representable and non-representable values.
- Sample addition example in the following format.

Floating Point Addition

X: _____ + Y: _____

X: _____

Y: _____

Step 1: Extract the exponent and value

Sign_x: _

Sign_y: _

Exp_x: _ _ _ _ _

Exp_y: _ _ _ _ _

Val_x: _____ * _____

Val_y: _____ * _____

2's Val_x: _____ * _____

2's Val_y: _____ * _____

Step 2: Equalize the exponents

Value of Largest Exp: _____

X: __ or Y: __

Shift Direction: _____ Shift Amount: _____

X: __ or Y: __

New Val_{__}: _____ * _____

X: __ or Y: __

Step 3: Integer addition

Val_x: _____ * _____

Val_y: _____ * _____

Val_R: _____ * _____

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Submission: Submit the final design to (use file name Lab2_RollNo.circ)

<https://u.pcloud.com/#/puplink?code=nA2XZxYB004m4SPF55SjMkHSdc5B55WpX>